LEMMA PURITY SMOOTH OVER DEPTH2 IN PIONE.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter pione.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:pione.tex:lemma-purity-smooth-over-depth2.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0BQF: sources/pione.tex, source lines 2019–2019. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BLR: sources/local-cohomology.tex, source lines 328–328. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0337: sources/algebra.tex, source lines 46374–46374. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EY6: sources/pione.tex, source lines 5220–5220. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original theorem 00FV: sources/algebra.tex, source lines 6524–6524. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07NE: sources/algebra.tex, source lines 44675–44675. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 057G: sources/more-morphisms.tex, source lines 10775–10775. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05D5: sources/algebra.tex, source lines 38407–38407. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00MB: sources/algebra.tex, source lines 22956–22956. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EYA: sources/pione.tex, source lines 5807–5807. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $U \subset X$ be an open subscheme. Assume*

(1) *f is smooth,*
(2) *S is Noetherian,*
(3) *for $s \in S$ with $\text{depth}(\mathcal{O}_{S,s}) \leq 1$ we have $X_s = U_s$,*
(4) *$U_s \subset X_s$ is dense for all $s \in S$.*

Then $F\acute{E}t_X \to F\acute{E}t_U$ is an equivalence.

Proof. The functor is fully faithful by Lemma 0BQF combined with Local Cohomology, Lemma 0BLR (plus an application of Algebra, Lemma 0337 to check the depth condition).

Let $\pi : V \to U$ be a finite étale morphism. Let $Y \to X$ be the finite morphism constructed in Lemma 0EY6. We have to show that $Y \to X$ is finite étale. To show that this is true for all points $x \in X$ mapping to a given point $s \in S$ we may perform a base change by a flat morphism $S' \to S$ of Noetherian schemes such that s is in the image. This follows from the compatibility of the construction in Lemma 0EY6 with flat base change.

After enlarging U we may assume $U \subset X$ is the maximal open over which $Y \to X$ is finite étale. Let $Z \subset X$ be the complement of U. To get a contradiction, assume

$Z \neq \emptyset$. Let $s \in S$ be a point in the image of $Z \to S$ such that no strict generalization of s is in the image. Then after base change to $\mathrm{Spec}(\mathcal{O}_{S,s})$ we see that $S = \mathrm{Spec}(A)$ with $(A, \mathfrak{m}, \kappa)$ a local Noetherian ring of depth ≥ 2 and Z contained in the closed fibre X_s and nowhere dense in X_s. Choose a closed point $z \in Z$. Then $\kappa(z)/\kappa$ is finite (by the Hilbert Nullstellensatz, see Algebra, Theorem 00FV). Choose a finite flat morphism $(S', s') \to (S, s)$ of local schemes realizing the residue field extension $\kappa(z)/\kappa$, see Algebra, Lemma 07NE. After doing a base change by $S' \to S$ we reduce to the case where $\kappa(z) = \kappa$.

By More on Morphisms, Lemma 057G there exists a locally closed subscheme $S' \subset X$ passing through z such that $S' \to S$ is étale at z. After performing the base change by $S' \to S$, we may assume there is a section $\sigma : S \to X$ such that $\sigma(s) = z$. Choose an affine neighbourhood $\mathrm{Spec}(B) \subset X$ of s. Then $A \to B$ is a smooth ring map which has a section $\sigma : B \to A$. Denote $I = \mathrm{Ker}(\sigma)$ and denote $B^\wedge$ the I-adic completion of B. Then $B^\wedge \cong A[[x_1, \ldots, x_d]]$ for some $d \geq 0$, see Algebra, Lemma 05D5. Observe that $d > 0$ since otherwise we see that $X \to S$ is étale at z which would imply that z is a generic point of X_s and hence $z \in U$ by assumption (4). Similarly, if $d > 0$, then $\mathfrak{m}B^\wedge$ maps into U via the morphism $\mathrm{Spec}(B^\wedge) \to X$. It suffices prove $Y \to X$ is finite étale after base change to $\mathrm{Spec}(B^\wedge)$. Since $B \to B^\wedge$ is flat (Algebra, Lemma 00MB) this follows from Lemma 0EYA and the uniqueness in the construction of $Y \to X$. $\square$